

mei

Metabolic Intelligence: Budget-Native Compute from Tag to Campus

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Measurement. Research study, not a journal final. Energy figures from the software reference are OpCounter analytical estimates, not board power. We have not synthesized or metered an FPGA board. No fabricated citations.

A budget goes in; the best answer that budget can buy comes out. Digital enzymes at microwatts; grid-following service at 100 MW. A post-transformer class: physics-informed, test-time, zero-shot capable, not pre-training required.

Abstract

Metabolic Intelligence is a budget-native class of computer intelligence. A stated energy budget goes in; the best answer that budget can buy comes out. The class is not a transformer update. It sits with Logical Intelligence (energy-based settle), World Labs (spatial world models), and other post-transformer programs as a next wave: intelligence whose first coordinate is the joule envelope, not the next-token corridor.

Class properties, stated plainly:

1. **Physics-informed.** Features and actuators respect sensor kinetics, heater physics, Landauer floors, and grid constraints as settled law, not optional add-ons.
2. **Test-time inference.** Binding, routing, and lever selection happen under the live budget; the decision is shaped at serve time.
3. **Super learning / sequential learning.** New contexts and classes can be absorbed under the same budget without a full pretrain replay.
4. **Zero-shot capable under grammar.** When Lookup or Formula covers the coordinate, the system answers without opening a generative leaf.
5. **Not pre-training required** as the default path. Abundance updates, family indices, and obtain-routers learn and decide under local cost; large corpus pretrain is residual, not the substrate.

Dual-scale proof in this study:

- **Tag scale.** Digital enzymes on cold-chain gas tags. Cortex-M4 instruction counts (emulator-measured): enzyme families with early exit take 1,723 instructions vs 2,451 for an int8 MLP at matched accuracy. On SmellNet sequential learning (50 foods, five groups, no revisit), enzymes keep 77.7% vs 18.4% for the MLP. A metabolic heating schedule draws about 10 μW total (datasheet + schedule model): heater about 3.0 μW , MCU sleep about 6.9 μW on STM32U3; about eight years on a CR2032. Classifier decisions are about 92 nJ; the sleep floor dominates on the metabolic schedule. Chicken-spoilage Anwar results are internal only and are **not published** here (no licence).
- **Campus scale.** A metabolic grid service for a 100 MW AI campus, tested end to end against a real OpenADR 3.1 VTN and a real MQTT DSX Flex broker. Plant is a model; protocols are real. With node sleep available: zero time over the grid cap across the closed scenario set. Headline lever order: pause deferral $\rightarrow$ slow per-user speed $\rightarrow$ route smaller model $\rightarrow$ lower effort $\rightarrow$ battery $\rightarrow$ shed.

Metabolic Intelligence (MEI once in this abstract) owns the **budget envelope and actuators** that make companion floors bind. [Mixture of Limits](#) chooses the gear. [Notational Intelligence](#) owns certify-before-commit. [Satiation](#) owns stop when Value of Information (Vol) is zero or budget refuse fires. This study owns sleep/heater schedules, OpenADR/MQTT service levers, and the (J(mlq)) obtain-router that prices look up $\rightarrow$ calculate $\rightarrow$ model $\rightarrow$ guess under λ at stake.

Hardware dual-map: the class runs today on GPUs and MCUs (von Neumann), and points at post-von Neumann substrates (neuromorphic, analog / in-memory, photonic, reversible / adiabatic) where energy-based and latent settle map natively. Estimates remain estimates. Soft-ref path: board_synth_claimed=false; package measured_j only when Metered.

1. Introduction: budget-native intelligence

People should not have to fight over energy to eat or to use AI. That civic sentence is the design brief. Cold-chain tags that watch food should last years on a coin cell. Campuses that serve answers should follow the grid when the grid asks them to cut, without inventing false board joules or collapsing every research law into one paper.

The dominant construction for computer intelligence still treats the transformer corridor as the substrate: pretrain a large generative model, then spend tokens and watts at serve time. That corridor has produced real products. It is not the only class. **Logical Intelligence** casts reasoning as energy minimization over constraints. **World Labs** casts spatial intelligence as world models that predict views and dynamics. **Metabolic Intelligence** casts intelligence as **budget-native obtain and actuate**: the envelope is first-class; the answer is the best the envelope can buy.

Shannon (1948) prices bits under uncertainty as settled law. Howard's Value of Information prices whether another observation is worth its cost for a decision. Landauer (1961) prices irreversible bit erasure in joules at temperature (T): ($E_{\text{bit}} = k_B T$) per bit erased in the ideal model. On receipts

that Landauer quantity is a **labeled estimate**, not a wattmeter reading. Together they imply floors: past a point, more bits stop buying outcomes that matter for a stated benefit. Metabolic Intelligence embodies those floors as **schedules and actuators** at the edge of a tag and at the meter of a campus.

Thesis. A budget goes in; the best answer that budget can buy comes out. The substrate is digital enzymes and other algorithmic compute under energy budgets, plus a grid-following service that pulls levers in order of least service lost per megawatt freed. Class properties follow: physics-informed features and heaters; test-time binding and routing; sequential / super learning without full pretrain replay; zero-shot close when grammar covers; pretrain as residual leaf, not default. Computers are hardware; software is applied engineering under constraints.

Three measurement facts constrain every later number. First, curve and ledger watt-hours are **model / measured-kernel / looked-up** composites, labeled by evidence class. Second, tag microwatts mix **datasheet** and **schedule model**; instruction counts are **emulator-measured**. Third, `board_synth_claimed=false`; package `measured_j` appears only when a labeled meter returns a reading.

2. Companion laws and OpenIE map

Four studies on this catalog compose. They do not collapse into one paper.

Role	Study	Owns
Navigation	Mixture of Limits	Which gear closes: Lookup → Formula → Solver → Model LAST. Floors: Vol, grammar, energy estimate, certificate, settle-refuse.
Commit	Notational Intelligence as Commit Law	propose → certify → commitrefuse → receipt.
Economic Reality of Satiation	Satiation and Scarcity after Free AI	Stop when Vol is zero on completeness (C(z)), or when budget / policy refuse fires.
Energy budget / metabolic embodiment	This paper (Metabolic Intelligence)	Budget envelope and actuators: tag sleep/heater; campus OpenADR/ MQTT levers; (J(m

flowchart TB

```
NAV["Navigation<br>Mixture of Limits"]
COM["Commit<br>Notational Intelligence"]
ECO["Economic Reality of Satiation"]
MET["Energy budget<br>Metabolic Intelligence"]
NAV --> COM --> ECO
MET -. -> |binds floors at tag and campus| NAV
```

MET → COM
MET → ECO

Mixture of Limits owns *which gear closes*. Notational Intelligence owns the irreversible commit shape. Satiation owns Economic Reality of Satiation. **Metabolic Intelligence** owns the envelope that makes those floors executable in joules and watts. Soft-ref path: board_synth_claimed=false; estimates ≠ measured_j.

OpenIE map

Surface	URL	Relation
Research	research.openie.dev	This study is /papers/mei/. Companions: Mixture of Limits, Notational Intelligence, Satiation.
Stack	stack.openie.dev	Family map; MEI is embodiment, not another stack card.
Compute	compute.openie.dev	Primitive table the cascade navigates; metabolic actuators spend primitives under μ on hardware H.
Synthesis	synthesis.openie.dev	Cost surface ($E(x)=(p)(p,H)$); this study extends obtain cost as ($J(m$
Verify	proof.openie.dev	Cousin energy law for verified artifacts.

Living placeholder for figures: [/living/mei/](https://living/mei/). Soft-ref path keeps package energy Estimated unless Metered.

3. Standing obtain rule and the ($J(mlq)$) synthesis router

Standing rule (David Charlot): **Don't calculate what can be looked up. Don't guess what can be calculated. Don't spend hard-guessing effort on an easy guess.**

The OpenIE synthesis surface prices computing an answer:

$$[E(x) = _ \{p P(c(x))\} (p)(p,H)]$$

Metabolic Intelligence raises resolution on *how each input is obtained*. For need (q) feeding decision (y), mechanism (m) has cost:

$$[J(mlq) = E_m + S_q^2 _m(t)]$$

$$[_m(t) = (_m^2 + _q t)(1 + 3p_m), m^* = _m J(mlq)]$$

Symbol	Meaning	Evidence
(E _m)	Energy of the obtain mechanism	Stack formula on the mechanism's primitives
(S _q)	Elasticity (dy / dq)	Calculated by central differences; never assumed
()	Joules at stake in the decision	Decision scale (one request; campus-year)
(_m , _q , t, p _m)	Uncertainty, staleness rate, age, zone-misjudge chance	Looked-up spread, derivation, or labeled guess

What falls out as law of the function, not as slogan:

- Lookups dominate when they measure the same quantity.
- Cheap crude options win when (S_q²) is small.
- Required precision: (^{*} =).
- Refresh a cache at (t^{*} = 2E / (S_q²)).

Ledger outputs (synthesis pack; math + sourced ledger; **not** board measured_j):

Output	Point	p10–p90	Measure next
Wh per V4 Pro answer at today's speed (GB300)	0.20 Wh	0.044–0.68	Meter one GPU pool for a day
Work-per-MW gain, speed matching human chat	1.92×	1.19–2.93	Pro speed; human share of tokens
Same gain on Vera Rubin	2.0×	1.2–3.4	Metered Rubin chat curve

Fused public proxies for output length (400 / 630 / 1,115 tokens) give 655 at log-σ 0.585, no tighter than a 500-token guess. The router keeps the guess for campus planning and takes the lookup per request. Only the gateway's own counters settle it. Evidence class: **model + looked-up + calculation**.

```

flowchart LR
    L[Look up] --> C[Calculate]
    C --> M[Model]
    M --> G[Guess]
    J["argmin J(m|q)"] --> L
    J --> C
    J --> M
    J --> G

```

4. Digital enzymes

A **digital enzyme** is a silicon recognition unit with a binding site, a product, an allosteric context switch, Hill cooperativity, an abundance that acts as the learned weight, and a cost for every binding check. A bit-enzyme variant uses no multiplies. The unit is algorithmic compute under a local energy cost, not a smaller transformer.

Result	Number	Evidence class	Where
Instructions / decision, Cortex-M4, 12 features	Enzymes + early exit 1,723; int8 MLP 2,451; first enzyme pool 6,948	Emulator-measured	mcu/
Accuracy, same test (3 seeds)	Enzymes 82.9 / 86.9 / 88.7%; MLP 82.2 / 78.0 / 87.4%	Real protocol on simulated features	benchmarks/
Late-life accuracy, physics-invariant features (days 90–180)	LDA 94.9%, MLP 93.4%, enzymes 90.0% (from 80.8%), Tsetlin 89.1%	Simulation + classifier	benchmarks/
SmellNet, 50 foods, 5 sequential groups, no revisit	Enzymes 77.7%; MLP 18.4%	Real dataset protocol	benchmarks/
SmellNet cross-day	MLP 85.5%; enzymes 79.8% (same on Cortex-M4)	Real dataset + MCU	benchmarks/

Class properties, grounded:

- **Physics-informed.** Drift-invariant / physics-informed features added about 9–11 points of late-life accuracy for enzymes, bit enzymes, and Tsetlin machines. Sensor kinetics and humidity/heater physics are first-class inputs, not post-hoc regularizers.
- **Sequential / super learning.** Enzymes keep usable accuracy while learning foods one group at a time without revisiting. The MLP collapses under the same protocol (77.7% vs 18.4%). That is the super-learning property under a budget: absorb new context without a full pretrain replay.
- **Test-time inference.** Binding checks, early exit through the family index, and allosteric context switches are serve-time decisions under abundance and cost.
- **Not pre-training required** as default. Abundance updates and family indices learn locally. Large generative pretrain is not the substrate of the tag classifier.
- **Zero-shot under grammar.** When a Lookup or Formula leaf covers the context (known context state, known refuse), the cascade closes without opening a generative Model leaf. Mixture of Limits states that order; enzymes supply a budget-native leaf when Model is residual.

Chicken-spoilage evaluation (Anwar & Anwar 2025) has **no licence**. Results from that dataset are **internal only, not published**. Soft-ref path: board_synth_claimed=false.

5. Cold-chain tag budget

On a gas tag the energy goes into **heating and sleep**, not into classifying. One gas reading costs about 0.5–27 mJ depending on the part (datasheet). A classifier decision costs about 92 nJ on an STM32U3 (CoreMark 15.5 $\mu\text{A}/\text{MHz}$ at 3.3 V; **calculated** from datasheet, not board package measured_j).

Metabolic schedule (about 9.6 readings/day), October 2026 parts:

Term	Power	Evidence
Heater (BME690 metabolic)	$\sim 3.0\ \mu\text{W}$	Datasheet + schedule model
MCU sleep (STM32U3)	$\sim 6.9\ \mu\text{W}$	Datasheet
Tag total	$\sim 10\ \mu\text{W}$	Sum of schedule model
CR2032 life	$\sim 8.1\ \text{years}$	Capacity model
Same tag on nRF54L15 sleep	$\sim 5.1\ \mu\text{W}$	Datasheet swap

Correction (honesty): earlier prose said “the heater is the cost.” True for frequent readings. On the metabolic schedule the **MCU sleep floor** is the larger term. Choose the MCU for sleep current. Innatera Pulsar-class neuromorphic parts publish audio/radar milliwatts, not gas-tag microwatt schedules; they are peers on the post–von Neumann map, not drop-in tag winners today.

Market timing (rules, not hype): FDA proposed FSMA 204 compliance move to 20 Jul 2028; Congress barred earlier enforcement. EU PPWR applies since 12 Aug 2026.

6. Campus energy per answer

Method (curves pack): energy per answer $\approx$ output tokens $\times$ all-in kW per GPU $\div$ output tokens/s per GPU.

- **Throughput:** NVIDIA AI Configurator 0.12.0 (measured GPU kernels; SGLang; NVFP4). Evidence: **measured-kernel model**.
- **Power per GPU from InferenceX** (tokens/s per chip $\div$ tokens/s per MW): GB300 2.12 kW, B300 1.90, B200 1.71, Vera Rubin 3.30, MI355X 2.09. Evidence: **calculated from published measurements**.
- **Served speeds (Artificial Analysis medians):** V4 Pro 79 tok/s/user, V4 Flash 125, Qwen3-VL 48.5. Evidence: **looked up**.

Workload (GB300 unless noted)	Energy	Evidence class
Chat, DeepSeek V4 Pro (2k in / 500 out)	0.32 Wh at 50 tok/s/user; 0.92 at 79; 4.57 at 159	Curves method
Synthesis ledger best estimate today	0.20 Wh (p10–p90 0.04–0.68)	Ledger fuse
Chat, V4 Flash	49 mWh at 50; 180 at 159	Curves
Agentic turn, V4 Flash (32k in, 28k cached / 1k out)	0.16 Wh at 96	Curves
Video clip, 8 s 720p (Wan 2.2)	390 Wh	Curves / model

Plateau levers (field + model, not board measured_j):

- Vera Rubin NVL72 moves the knee of the speed-energy curve from about 100 to about 175 tok/s/user (InferenceX / MLPerf; **measured** field reports). Speed matching human chat is worth about 2.0× on Rubin vs about 2.9× on GB300 (ledger).
- Flash-first routing (V4.1 Flash answering most human chat) plus speed matching: about 2.3–4.4× more work per MW (efficiency pack; **model + looked-up**).
- Deeper speculative decoding (accept length 2.8 modeled): about 1.55× cut in today’s base-mix energy per output token; speed matching then adds only about 1.4–1.8×.

These are **test-time** levers under a live budget: which model answers, how fast it streams, how much draft speculation to accept. They are Metabolic Intelligence at campus scale, not a new pretrain run.

7. Metabolic grid service

The service makes a 100 MW AI campus follow grid caps. Lever order (least service lost per MW freed): **pause deferrable work** → **slow per-user speed** → **route to a smaller model** → **lower reasoning effort** → **battery** → **shed**.

Architecture (datacenter/service/):

- **Grid in:** OpenADR 3.1 VEN (OAuth2, polling, DEMAND reports), tested against real openleadr-rs VTN. Evidence: **real protocol**.
- **Backstop:** DSX Flex ISV over MQTT, CloudEvents, JWS ES256. Evidence: **real broker + schemas**.
- **Actuators:** Dynamo planner GPU budgets; metabolic gateway (in-flight caps, routing, effort, x-consumer speed tiers); Kueue / SGLang pause; Redfish node sleep; battery PCS.
- **Learning:** per-pool speed curve by recursive least squares; power-model bias loop (learned +5 MW on a hot day in the plant model).
- **Plant:** campus model. Evidence: **model**. Protocols and schemas: **real**.

Scenario (100 MW GB300 campus)	Time over cap	Interactive served	Note
A–C: 60/70% day; notice / feedforward / no notice	0	100%	Notice mostly saves battery
D: 90/95%, fast campus (159 tok/s/user)	0	100%	All knowledge chat routed; effort low
E: 90/95% without node sleep	96 min	1%	Idle GPUs alone exceed a 10 MW cap
L–M: today’s looked-up speeds, 60/70 and 90/95	0	100% / 52.6%	Speed already spent cannot be spent again
N: no notice, no battery (PJM-style 10 min)	7.5 min	99.8%	Under cap within 225 s
O: 60% cut for 10 h	0	100%	Deadlines kept
P: new cap every 5 min for 3 h	0	98.8%	ERCOT-style
Q: recovery ramp 2 MW/min	0	100%	Max rise 2.9 MW/min vs 29–47 unlimited
R: 90% cut for 10 h	0	13%	Video deadline vs interactive (policy)

In every run: no actuator call failed; VTN accepted all OpenADR reports; no DSX message rejected. Headline: **zero time over the grid cap whenever node sleep is available**. Correction honesty: “100% served through a 90/95% cut” assumed a fast campus and one curve; with per-pool curves it is about 94%, from today’s speeds about 53%. The **cap held** in all of them.

8. SOTA at the metabolic frontier

8.1 Class definition

Metabolic Intelligence is a budget-native / energy-envelope intelligence class: the system takes an explicit joule or power budget and returns the best obtainable answer under that envelope, rather than maximizing quality under uncapped transformer FLOPs. Architecturally it sits with Logical Intelligence–style moves *past* basic transformers (energy-based scoring and constraint commit, latent world models, non-autoregressive or non-attention primitives), but binds the envelope at **two scales at once** (microwatt tags [?] megawatt campuses). The unit of computation is the **digital enzyme** (recognition, allosteric context gating, abundance as weight, local learning and decay, costed binding check) and related algorithmic compute, not ever-larger nets.

Class properties (stated as operating claims; peer overlap cited honestly below):

Property	MEI grounding in this pack	Honest peer overlap
Super / sequential learning	SmellNet enzymes 77.7% vs MLP 18.4% across five groups without revisit (real protocol)	AdaTM continual Tsetlin; classical Hopfield / DenseAM pattern add. Do not claim universal end of catastrophic forgetting.
Test-time inference	Binding, early exit, (J(m	q)) route, campus lever dial under live budget
Zero-shot under grammar	Lookup → Formula → Solver first (Mixture of Limits); refuse when covered	V-JEPA 2-AC zero-shot robot planning without task reward is a strong peer. OpenADR demos here are protocol-tested , not “zero-shot AGI.”
No large pretrain required	Enzyme / bit-enzyme online path on MCU	Tsetlin / Hebbian AM cousins. Do not attribute this to JEPA, World Labs, or Logical Intelligence frontier stacks; they pretrain heavily. Scope the claim to the enzyme path.
Physics-informed	E-nose kinetics, drift-invariant features (+9–11 late-life points); heater / sleep schedule; Landauer as labeled estimate	PINN lineage; analog EBM as physics substrate. Physics features ≠ thermodynamic optimality without meters.

Unique OpenIE gap today: tag[?]campus dual scale + OpenADR 3.1 / DSX Flex–shaped service + (J(mlq)) obtain-router + digital enzymes + Satiation stop. Peers own slices. None presently combine that stack as one class definition. Soft-ref path: board_synth_claimed=false.

8.2 Method taxonomy (must-cite)

Method	vs transformer	Existing HW	Post–von Neumann	Ev.	MEI overlap
Certify/commit EBM (Logical Intelligence)	Next-token → energy landscape + Lean-checkable proofs	GPU + Lean; Kona / Aleph pilots (Jan 2026)	Analog DenseAM / Hopfield-style CIM (research)	V/C	Test-time verify. Constraint energy ≠ campus joules. Not “no pretrain.”
Latent JEPA / world models	Pixel/token gen → predict in latent; plan by latent energy	GPU ViT + MPC (V-JEPA 2-AC, Jun 2025); World Labs Marble / Atlas	Speculative: latent dynamics on analog associative recall	M/V	Zero-shot planning peer. Requires large video pretrain — conflicts with enzyme “no pretrain”

Method	vs transformer	Existing HW	Post-von Neumann	Ev.	MEI overlap
					unless claims are split by path.
DenseAM / Hopfield EBM	Attention cast as energy flow; retrieval = minimize E	Digital solvers on GPU/CPU	Analog RC + crossbar; memristor Hopfield (2026)	P/S	Test-time energy descent; physics-native dynamics
Thermodynamic / equilibrium EBM HW	Sampling Z digitally → Langevin/ Gibbs on physics	Superconducting / stochastic analog research	Hardware-native EBM sampling	P	Physics-informed substrate; early
Digital enzymes (OpenIE)	Dense MLP → recognition + abundance + budgeted binding	MCU / Cortex-M4 (emulator-measured instr.)	Content-addressable / sparse associative fit	M/S	Sequential learning; budget dial; no large pretrain on enzyme path; physics-informed features
Tsetlin / bit-logic	Real-valued nets → propositional clauses	Digital ASIC 65 nm (8.6 nJ/ MNIST frame, Jan 2025)	Stays digital-sparse	M/P	Online / continual peers; nJ edge
HDC + SDM	Dense embeddings → hypervectors / sparse distributed memory	SRAM/FPGA HDC; HDStream 2026	Associative CAM / memristive bundling	P/V	Fast associative recall; not the campus story
SSM / Mamba / RWKV	Quadratic attn → linear state	GPU; FPGA/ASIC	Still digital accelerators	M/P	Corridor efficiency peer only; not budget-native class
Photonic / analog CIM / adiabatic	Data-move → in-memory / charge recovery	Lab photonic fJ/ op claims; NeuRRAM lineage; Innatera Pulsar ~0.4–0.6 mW audio/radar	Core post-vN joule-floor path	M/V/S	Joule floor hardware; not the MEI software class by itself

Method	vs transformer	Existing HW	Post-von Neumann	Ev.	MEI overlap
Power-aware serving (soft)	Same transformer + power caps / speed dial	Dynamo 1.5 (18 Sep 2026); DPS	N/A	V/M	Illustrates budget binding at campus; not a new intelligence class
Grid-flexible campus (soft)	Flat load → dispatchable envelope	DSX Flex; OpenADR 3.x; AEMA (16 Sep 2026, no spec yet)	N/A	C/V/R	Envelope as grid budget; OpenIE wires this to obtain-router

8.3 Positioning vs named peers

Peer	Their axis	Metabolic Intelligence difference
Logical Intelligence	Constraint energy + formal commit	Joule envelope + obtain routing + dual scale
World Labs / V-JEPA	Spatial / latent world models	Budget-native enzymes + campus grid service; latent optional
DenseAM / analog EBM	Physics dynamics for inference	Software enzyme class plus envelope ops; cite as HW future
Dynamo / DSX / AEMA	Power and grid actuators	Actuators inside Mixture of Limits intelligence
Tsetlin / HDC	Efficient non-transformer silicon	Cousins at edge; lack campus (J(m

```

flowchart TB
  subgraph classwave [Post-transformer class wave]
    LI[Logical Intelligence<br/>constraint EBM]
    WL[World Labs / JEPA<br/>latent worlds]
    MEI[Metabolic Intelligence<br/>budget envelope]
  end
  subgraph hw [Hardware dual-map]
    VN[GPU / MCU today]
    PVN[Neuromorphic / analog CIM<br/>photonic / adiabatic]
  end
  MEI --> VN
  MEI --> PVN
  LI -. -> PVN
  WL --> VN

```

8.4 Soft peers only: serving and grid actuators

Dynamo 1.5 power-aware scaling, NVIDIA DPS, InferenceX Vera Rubin tok/s/MW curves, Emerald DSX Flex, and the AI Energy Management Alliance are **envelope actuators and field measurements**. They do not define the intelligence class. Academic joule-aware routers (PEARL, OmniRouter, Energy-Aware LRM) price which LLM answers; they do not price which obtain zone (Lookup → Formula → Solver → Model LAST) and do not bind tag[?]campus.

Term hunt (2 Oct 2026): **Metabolic Intelligence** as a named peer class to Logical Intelligence / World Labs was **not found** outside OpenIE / Charlot Lab energy-first framing. That is a positioning fact, not a monopoly claim on every energy paper.

8.5 What not to claim without meters

- Wh/answer, J/token, μ W lifetime, or $\times$ work/MW untied to pack evidence **M/C** or a named external meter.
- “Routing costs no quality” (Flash leads AA composite; trails ~ 1.5 GPQA).
- “Heater is always the tag cost” (sleep floor dominates on metabolic schedule).
- Rubin “up to 30 $\times$ ” without speed condition (gain is speed-dependent).
- AEMA / DSX Flex latency or ramp numbers (unspecified).
- Shipping neuromorphic gas/odour nJ figures (not found).
- That Logical Intelligence “energy-based” equals grid joules (different energy).
- That MEI needs no pretrain **and** matches V-JEPA zero-shot robotics (incompatible scopes; split by path).
- Chicken-spoilage accuracy in public prose (**no licence**).

Soft-ref path: board_synth_claimed=false; estimates $\neq$ measured_j.

9. Corrections log

Condensed from the Metabolic Intelligence handoff corrections log. Withdrawn claims are stated plainly.

Date	Withdrawn or revised	Correction
2 Oct 2026	FSMA 204 “compliance moved to 20 Jul 2028” as settled	FDA proposed ; Congress barred enforcement before that date
2 Oct 2026	PJM IRAS “full reduction within 10 min”; effective date 12 Oct 2026	Transmission owner must start and complete reduction within 10 min of PJM quantity; effective date unverifiable, removed
2 Oct 2026	STM32U3 at 12 μ A/MHz as CoreMark	That figure is a while(1) loop at 48 MHz; CoreMark is 15.5 μ A/MHz (51 pJ/cycle; 92 nJ/decision calculated)

Date	Withdrawn or revised	Correction
1 Oct 2026	“V4.1 Flash routing costs no quality”	Leads V4 Pro on AA composite (39 vs 36); trails ~1.5 GPQA points on knowledge
1 Oct 2026	Recovery ramp as fixed rising line	Lagging actuators jumped 13–31 MW/min; redesigned plan+one-step with battery on meter (2.9 MW/min)
1 Oct 2026	100% served through 90/95% cut as universal	Fast campus / one curve; per-pool curves ~94%; today’s speeds ~53%; cap held in all
1 Oct 2026	Speed-matching gain 2–4× / 4.6–12×	Baselines were guesses; looked-up speeds → 1.65–2.9×
1 Oct 2026	All-in kW 2.25 / 2.0 / 2.15 (GB300/B300/B200)	InferenceX-derived: 2.12 / 1.90 / 1.71
1 Oct 2026	B300 “as good per Wh at low speed”	Withdrawn; GB300 2.9× B300 tokens/MW at 96 tok/s/user (InferenceX)
1 Oct 2026	Chat answer 5–62 mWh from InferenceX J/token	InferenceX counts total tokens (mostly cached input); replaced by AI Configurator curves
1 Oct 2026	Flash at 0.33× Pro energy/token	AI Configurator: 0.04–0.15×; routing stronger, speed dial weaker
1 Oct 2026	Edge: “heater is the cost”	Frequent readings yes; metabolic schedule → MCU sleep floor dominates
Sep 2026	Enzyme MCU cost estimated in Python	Real counts 2.3–2.9× MLP; family index → 0.70×
Sep 2026	Older systems as current SOTA; XOR as two-site MWC	Relabelled or removed

Chicken-spoilage Anwar numbers stay **unpublished** (no licence).

10. Measurement honesty and open items

Evidence classes used in this study: **real protocol**, **emulator-measured**, **datasheet**, **measured-kernel model**, **looked-up**, **calculation**, **plant model**, **estimate**. Estimates ≠ measured_j. Soft-ref path: board_synth_claimed=false.

Ranked open measurements (synthesis ledger stakes):

1. Meter one GB300 pool for a day (narrows energy-per-answer band; campus-scale Vol of the meter).
2. Vera Rubin chat curve from a metered pool or AISimulate once Rubin-supported.
3. Gateway counters: output tokens per answer; human share (x-consumer).
4. V4.1 Flash throughput curve (replace V4 Flash stand-in behind routing).
5. Speculative-decoding accept length per workload from engine telemetry.

Service gaps: backup-generator start (PJM 15 min); trip with no control signal then controlled restart; ride-through (MISO / NERC due 31 Dec 2026); Dynamo 1.5 / DPS as live power-cap actuators; operator setting for “deadlines ahead of interactive.”

Edge gaps: SGP41 single-shot validation; radio uploads and cold-temperature battery; low-power sensor data (SmellNet uses MQ-series parts). Admin: licence for code pack; chicken-dataset authors’ permission before any derived publish.

Stage C / package measured_j for Metabolic Intelligence hardware remains **open**. Tag figures are datasheet + schedule + emulator. Campus figures are curves + plant model + real protocols. No invented board package joules.

11. Conclusion and relation to Mixture of Limits product

Metabolic Intelligence teaches a budget-native class: physics-informed, test-time, zero-shot under grammar, sequential / super learning without mandatory pretrain, dual-mapped to GPUs/MCUs today and to neuromorphic / analog / photonic / adiabatic substrates next. A budget goes in; the best answer that budget can buy comes out. Digital enzymes prove the class at microwatts. The OpenADR / DSX Flex service proves the class at 100 MW. The (J(mlq)) router makes the standing obtain rule executable.

Companions stay distinct. [Mixture of Limits](#) navigates Lookup → Formula → Solver → Model LAST and owns the product path (mol.yaml / mol run / A1–A14; [openIE-dev/mixture-of-limits](#)). [Notational Intelligence](#) owns certify-before-commit. [Satiation](#) owns Economic Reality of Satiation. This study owns the envelope and actuators that make those floors bind when food is watched on a coin cell and when a campus must shed watts without lying about meters.

Product surface under Mixture of Limits

Metabolic Intelligence does not replace the Mixture of Limits product path. It supplies the envelope those floors spend: the Mixture of Limits cascade decides the gear; metabolic actuators and the obtain-router decide how hard the envelope may be driven at tag sleep current and at campus import meters. When Vol is zero or the budget refuse fires, Satiation and Mixture of Limits already name stop as success. This study shows that stop is executable with OpenADR reports and node sleep, and that enzyme recognition can stay inside a coin-cell envelope without a transformer pretrain.

Soft-ref path: board_synth_claimed=false; estimates $\neq$ measured_j. Living figures: [/living/mei/](#) (placeholder; companions pending). PDF: [/pdfs/mei.pdf](#) when regenerated.

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